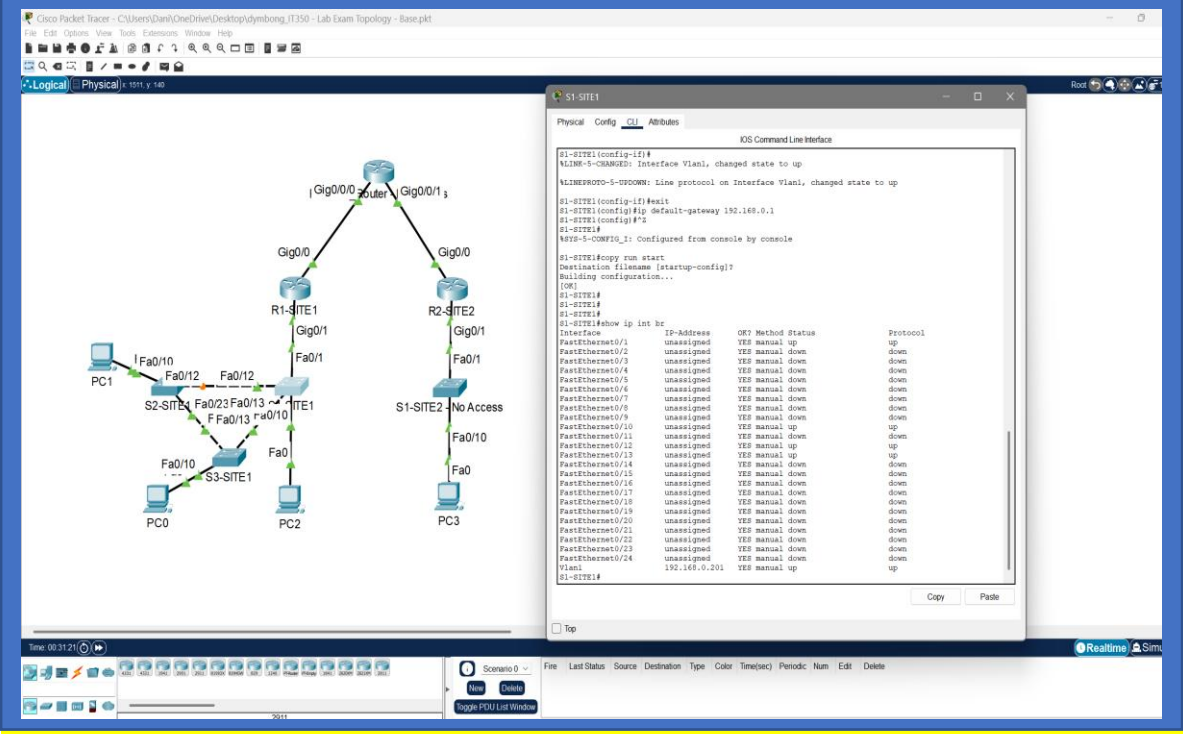


IT 350 – Final Exam – Lab Practical

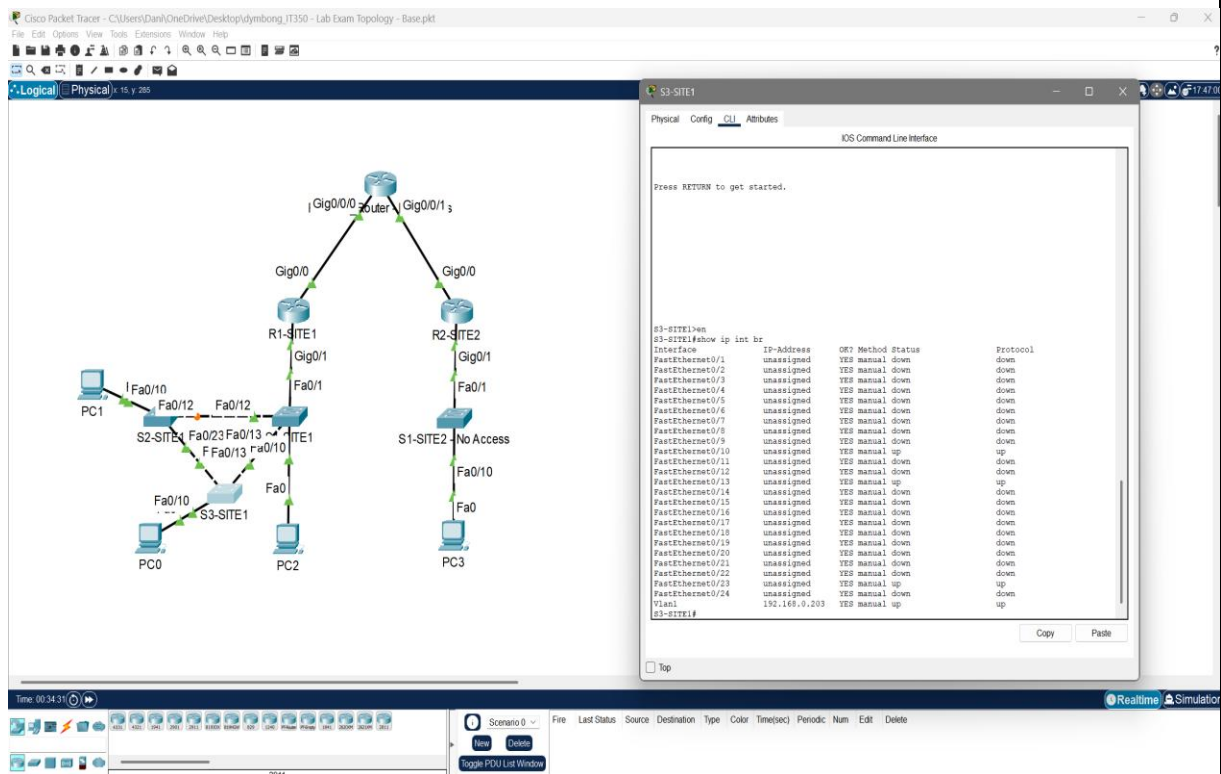
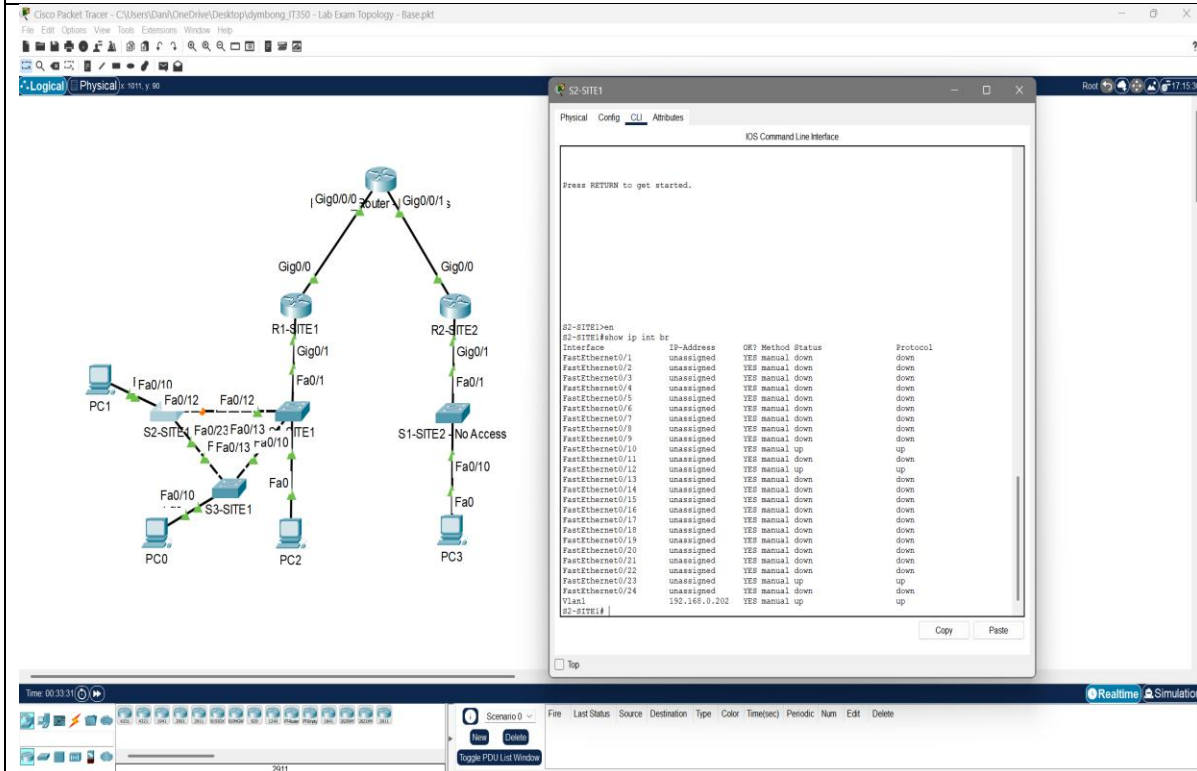
You have been provided with a two “site” topology connected by a provider router. You will not have, nor will you need, access to the provider router or to S1-SITE2. Various PCs are connected to the network. Complete each of the steps below to satisfy all requirements and achieve full connectivity between all PCs in the network.

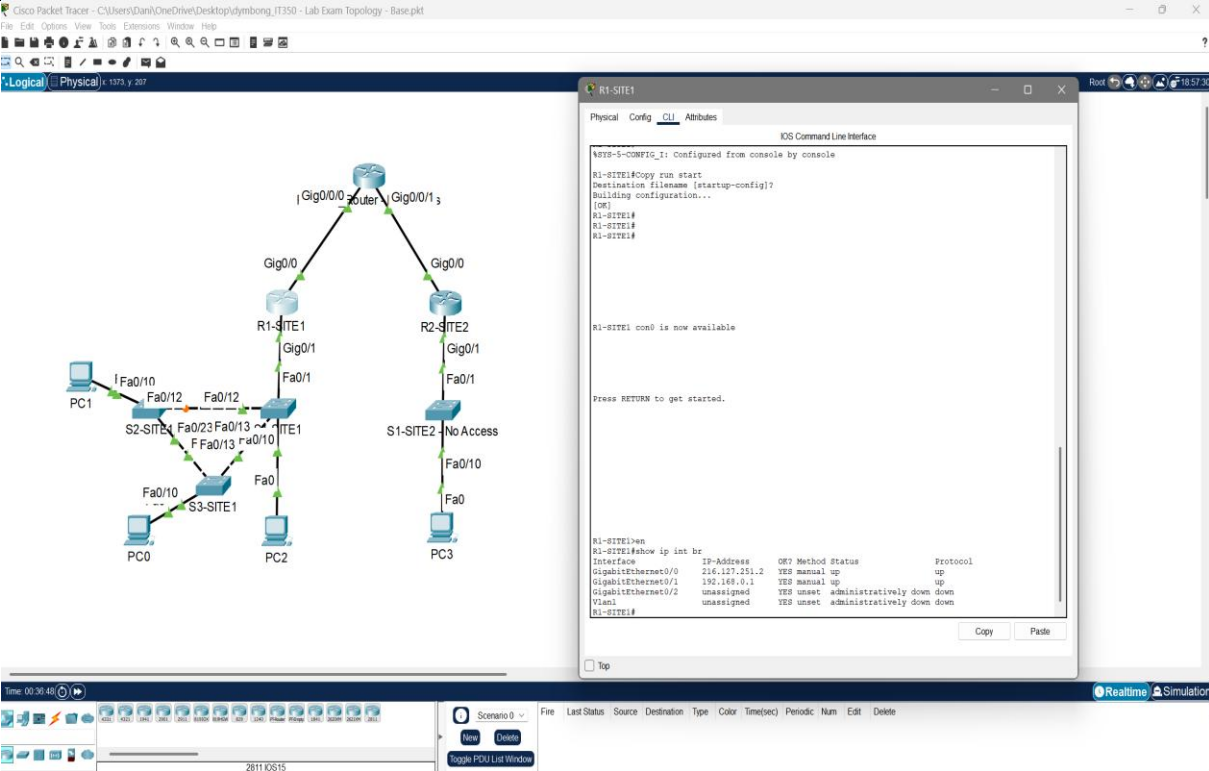
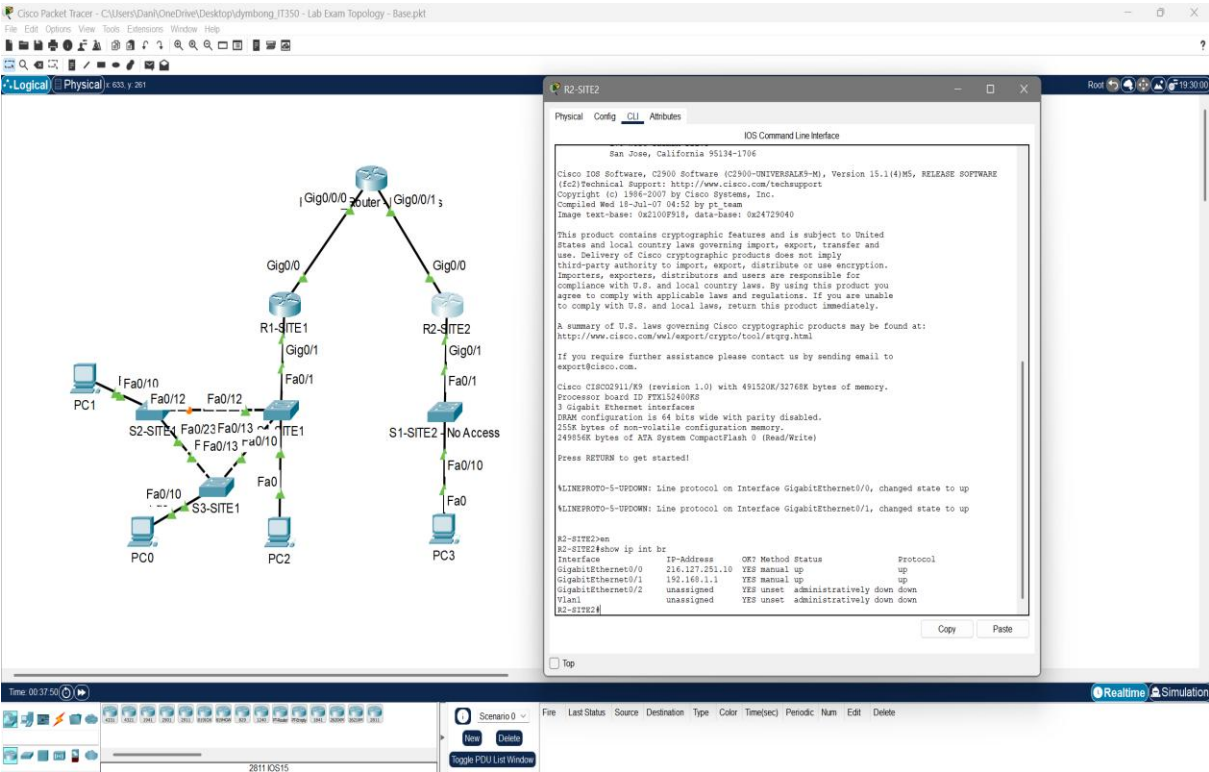
	Points Possib le	Points Award ed
1. Configure each of your routers and switches (NOT Provider Router, S1-SITE2, or the PCs) with a hostname. The hostname of each device should match the description label on the topology pallet. For example, the device labeled as S2-SITE1 on the topology view of PacketTracer should have a hostname of S2-SITE1. Add a screenshot of the output of the “show cdp neighbor” command from S1-SITE1 to show that you have completed this.	8	
2. Ensure that all interfaces are up at layer 1 and 2 so that you have ‘green’ status on all devices. Add a screenshot of the logical view of the Packet tracer window showing all interfaces with a green status.	8	

	Points Possib le	Points Award ed
3. Some addressing has already been configured for you. Some will need to be added. Configure IP addressing on your routers and switches as indicated below. Only use what you need and do not configure addressing that is not necessary: a. R1-SITE1 is connected to the provider router. The provider has given you the subnet of 216.127.251.0/30. The provider is using the first available host address in this range. Configure an address for your interface, but DO NOT use an address the provider has selected. b. All switches at SITE1 (S1-SITE1, S2-SITE1, and S3-SITE1) require an IP address for management. The subnet for this site is 192.168.0.0/24. Configure each VLAN1 interface on the switches with a usable IP address from the range. You can use any host address above .200 for this purpose. Switches should have full IP connectivity to other devices. Configure routes or default gateways as appropriate. c. Add a screenshot of the “show ip int brief” command for each of the switches/routers that you configured.		
 The screenshot displays a Cisco Packet Tracer workspace. On the left, a network topology is visible with a central 'Router' connected to three sites: R1-SITE1, R2-SITE2, and S1-SITE2. R1-SITE1 is connected to S2-SITE1 and S3-SITE1. S2-SITE1 and S3-SITE1 are connected to each other. PC1 is connected to S2-SITE1, and PC2 is connected to S3-SITE1. A 'No Access' label is present near S1-SITE2. On the right, a terminal window for R1-SITE1 shows the following configuration: <pre>R1-SITE1(config-if)# VLINE-S-CHANGED: Interface Vlan1, changed state to up VLINEPROTO-S-UPDOWN: Line protocol on Interface Vlan1, changed state to up R1-SITE1(config-if)#exit R1-SITE1(config)#ip default-gateway 192.168.0.1 R1-SITE1(config)# R1-SITE1# R1-SITE1#show ip int br Interface IP-Address OK? Method Status Protocol FastEthernet0/1 unassigned YES manual up up FastEthernet0/2 unassigned YES manual down down FastEthernet0/3 unassigned YES manual down down FastEthernet0/4 unassigned YES manual down down FastEthernet0/5 unassigned YES manual down down FastEthernet0/7 unassigned YES manual down down FastEthernet0/8 unassigned YES manual down down FastEthernet0/9 unassigned YES manual down down FastEthernet0/10 unassigned YES manual up up FastEthernet0/11 unassigned YES manual down down FastEthernet0/12 unassigned YES manual up up FastEthernet0/13 unassigned YES manual up up FastEthernet0/14 unassigned YES manual down down FastEthernet0/15 unassigned YES manual down down FastEthernet0/16 unassigned YES manual down down FastEthernet0/17 unassigned YES manual down down FastEthernet0/18 unassigned YES manual down down FastEthernet0/19 unassigned YES manual down down FastEthernet0/20 unassigned YES manual down down FastEthernet0/21 unassigned YES manual down down FastEthernet0/22 unassigned YES manual down down FastEthernet0/23 unassigned YES manual down down FastEthernet0/24 unassigned YES manual down down Vlan1 192.168.0.201 YES manual up up R1-SITE1#</pre>	8	

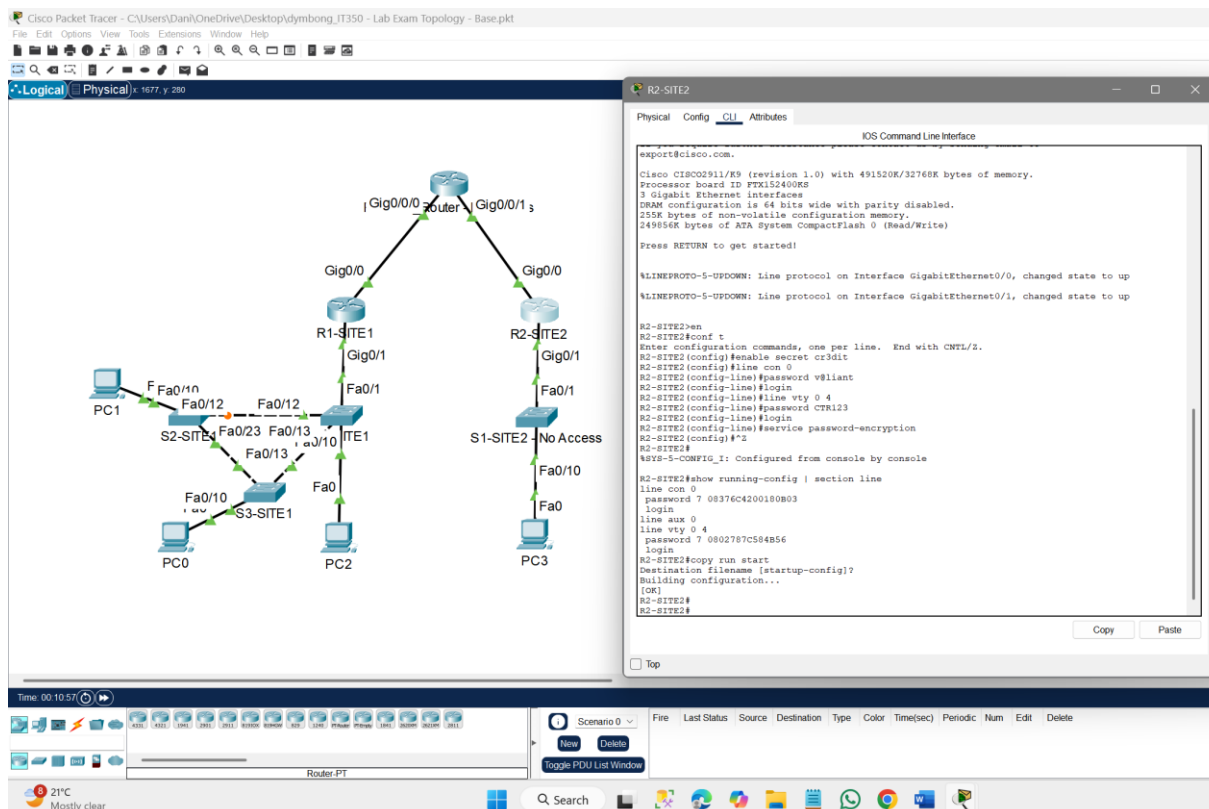
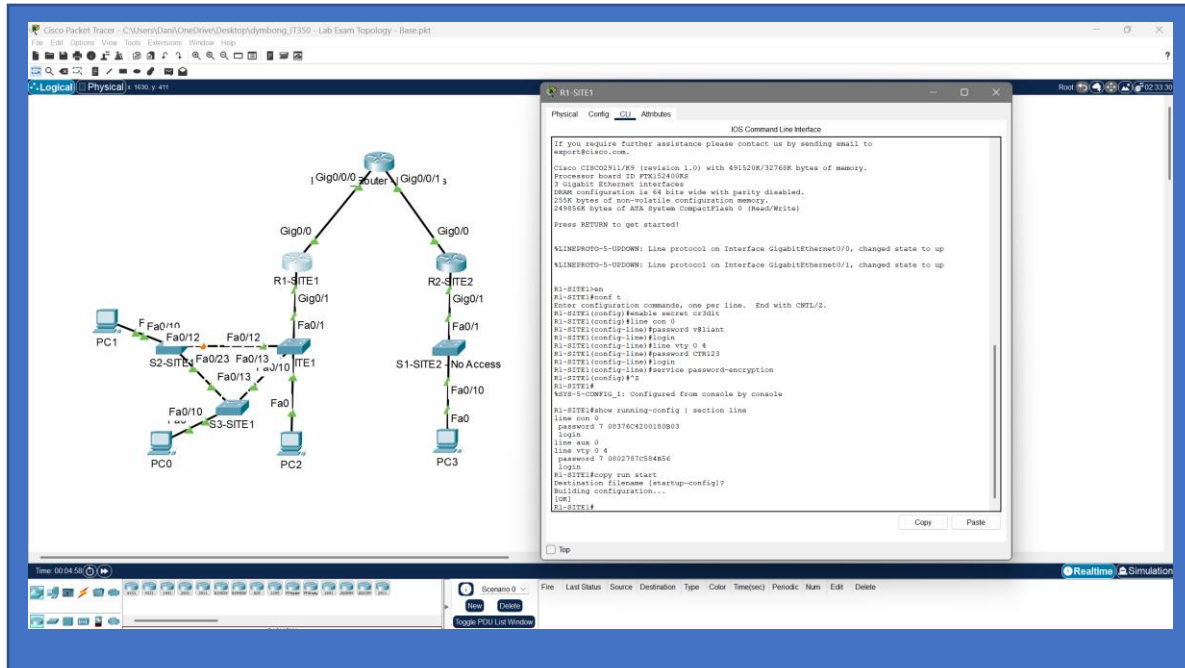
Points
Possib
le

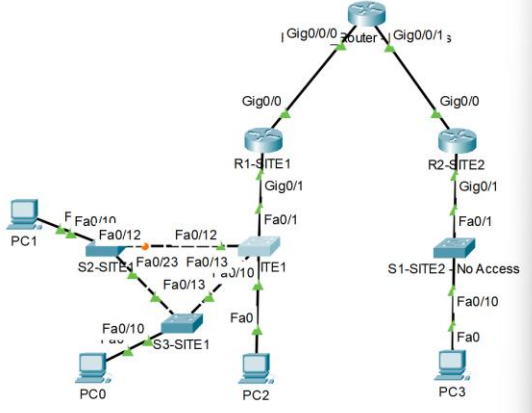
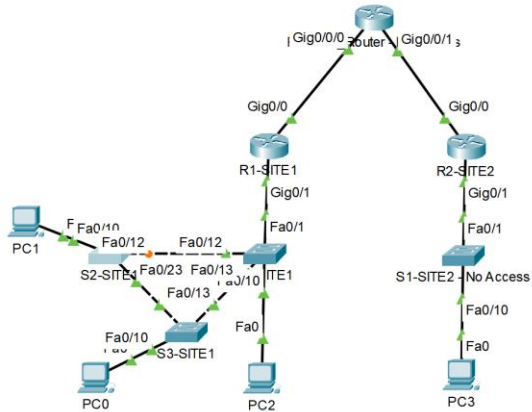
Points
Award
ed



	Points Possible	Points Awarded
 The screenshot shows a network topology in Cisco Packet Tracer. A central router is connected to three sites: R1-SITE1, R2-SITE2, and R3-SITE3. R1-SITE1 is connected to the central router via GigabitEthernet0/0/1. R2-SITE2 is connected via GigabitEthernet0/0/1. R3-SITE3 is connected via GigabitEthernet0/0/1. The topology also includes PCs (PC0, PC1, PC2, PC3) and switches (S1-SITE1, S2-SITE2, S3-SITE3). The CLI of R1-SITE1 is open, showing the configuration of the GigabitEthernet0/0/1 interface with IP address 192.168.0.1 and status 'administratively down'.		
 The screenshot shows the same network topology as above. The CLI of R2-SITE2 is open, showing the configuration of the GigabitEthernet0/0/1 interface with IP address 192.168.0.1 and status 'administratively down'.		
4. You would like to implement a simple method of locally and remotely accessing your devices and to discourage unauthorized access. Ensure that the following is configured on each device: - Configure all your devices with an enable password. The password is cr3dit. The password should NOT be visible in the show run and should ALWAYS be hashed. - Configure your console password to be v@liant. This password DOES NOT need to be hashed in the show run.		

- Enable telnet on all your devices. Ensure that a password is required for access. The password to use for telnet access is **CTR123**. This password DOES NOT need to be hashed. Telnet should allow up to five simultaneous connections when you are done configuring.
- Add a screenshot of the output of the command “show run | section line” from each of the devices you configured.



	Points Possib le	Points Award ed
Cisco Packet Tracer - C:\Users\Dan\OneDrive\Desktop\dybong.IT350 - Lab Exam Topology - Base.pkt File Edit Options View Tools Extensions Window Help Logical Physical 1486 y 264  PC1 PC2 PC3 R1-SITE1 R2-SITE2 S1-SITE1 S2-SITE1 S3-SITE1 Router Physical Config CLI Attributes IOS Command Line Interface <pre>%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to up %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to up %LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up %LINK-5-CHANGED: Interface FastEthernet0/13, changed state to up %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/13, changed state to up %LINK-5-CHANGED: Interface FastEthernet0/12, changed state to up %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/12, changed state to up S1-SITE1>en S1-SITE1#conf t Enter configuration commands, one per line. End with CNTL/Z. S1-SITE1(config)#enable secret cr3dit S1-SITE1(config)#line con 0 S1-SITE1(config-line)#password v8llant S1-SITE1(config-line)#login S1-SITE1(config-line)#line vty 0 4 S1-SITE1(config-line)#password CT8123 S1-SITE1(config-line)#login S1-SITE1(config-line)#service password-encryption S1-SITE1(config)#^Z S1-SITE1# %SYS-5-CONFIG_I: Configured from console by console S1-SITE1#show running-config section line line con 0 password 7 08376C4200100803 login line vty 0 4 password 7 0802787C584B56 login line vty 5 15 login S1-SITE1#copy run start Destination filename [startup-config]? Building configuration... [OK] S1-SITE1#</pre> Time: 00:14:30 Router-PT Cisco Packet Tracer - C:\Users\Dan\OneDrive\Desktop\dybong.IT350 - Lab Exam Topology - Base.pkt File Edit Options View Tools Extensions Window Help Logical Physical 690 y 403  PC1 PC2 PC3 R1-SITE1 R2-SITE2 S1-SITE1 S2-SITE1 S3-SITE1 Router Physical Config CLI Attributes IOS Command Line Interface <pre>%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to up %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to up %LINK-5-CHANGED: Interface FastEthernet0/23, changed state to up %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up %LINK-5-CHANGED: Interface FastEthernet0/12, changed state to up %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/12, changed state to up S2-SITE1>en S2-SITE1#conf t Enter configuration commands, one per line. End with CNTL/Z. S2-SITE1(config)#enable secret cr3dit S2-SITE1(config)#line con 0 S2-SITE1(config-line)#password v8llant S2-SITE1(config-line)#login S2-SITE1(config-line)#line vty 0 4 S2-SITE1(config-line)#password CT8123 S2-SITE1(config-line)#login S2-SITE1(config-line)#service password-encryption S2-SITE1(config)#^Z S2-SITE1# %SYS-5-CONFIG_I: Configured from console by console S2-SITE1#show running-config section line line con 0 password 7 08376C4200100803 login line vty 0 4 password 7 0802787C584B56 login line vty 5 15 login S2-SITE1#copy run start Destination filename [startup-config]? Building configuration... [OK] S2-SITE1#</pre> Time: 00:17:53 Router-PT Cisco Packet Tracer - C:\Users\Dan\OneDrive\Desktop\dybong.IT350 - Lab Exam Topology - Base.pkt File Edit Options View Tools Extensions Window Help		

Cisco Packet Tracer - C:\Users\Dan\OneDrive\Desktop\JIT350 - Lab Exam Topology - Base.pkt

Logical Physical 429, y 97

S3-SITE1

Physical Config CLI Attributes

IOS Command Line Interface

```

VLINE-5-CHANGED: Interface FastEthernet0/10, changed state to up
VLINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to up
VLINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
VLINE-5-CHANGED: Interface FastEthernet0/23, changed state to up
VLINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up
VLINE-5-CHANGED: Interface FastEthernet0/13, changed state to up
VLINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/13, changed state to up

S3-SITE1#en
S3-SITE1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S3-SITE1(config)#enable secret c3d1t
S3-SITE1(config)#line con 0
S3-SITE1(config-line)#password v8llant
S3-SITE1(config-line)#login
S3-SITE1(config-line)#line vty 0 4
S3-SITE1(config-line)#password CTR123
S3-SITE1(config-line)#login
S3-SITE1(config-line)#service password-encryption
S3-SITE1(config)#
S3-SITE1(config)#*
S3-SITE1#
APV2-5-CONFIG_I: Configured from console by console

S3-SITE1#show running-config | section line
line con 0
password 7 08376C4200180B03
login
line vty 0 4
password 7 0802787C584B56
login
line vty 5 15
login
S3-SITE1#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
S3-SITE1#
  
```

5. Create a new VLAN 20 in your switch environment at SITE1 and name it 'DEV'. Add it as an access port on the interface connected to PC0 connected to the S3-SITE1 switch. Ensure that this VLAN is available on all switches by whatever means you prefer. **NOTE: when you do this, PC0 will lose IP connectivity to other devices. This issue will be resolved in subsequent steps.** Add a screenshot of the output of the "show vlan" and "show int trunk" commands for each of the three switches.

Cisco Packet Tracer - C:\Users\Dan\OneDrive\Desktop\JIT350 - Lab Exam Topology - Base.pkt

Logical Physical 429, y 97

S1-SITE1

Physical Config CLI Attributes

IOS Command Line Interface

```

Building configuration...
[OK]
S1-SITE1#
S1-SITE1#show interfaces trunk
Port      Mode      Negotiation Status    Native VLAN
Fa0/21    desirable on      802.1q    trunking   1
Fa0/22    desirable on      802.1q    trunking   1
Fa0/23    on        802.1q    trunking   1
Fa0/24    on        802.1q    trunking   1

Port      VLANs allowed on trunk
Fa0/21    1-1005
Fa0/22    1-1005
Fa0/23    1-1005
Fa0/24    1-1005

Port      VLANs allowed and active in management domain
Fa0/21    1,20
Fa0/22    1,20
Fa0/23    1,20
Fa0/24    1,20

Port      VLANs in spanning tree forwarding state and not pruned
Fa0/21    1,20
Fa0/22    1,20
Fa0/23    1,20
Fa0/24    1,20

S1-SITE1# show vlan brief
VLAN Name                Status    Ports
-----
1  default              active    Fa0/2, Fa0/3, Fa0/4, Fa0/5,
                                           Fa0/6, Fa0/7, Fa0/8, Fa0/9,
                                           Fa0/11, Fa0/14, Fa0/15, Fa0/16,
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
20  DEV                  active    Fa0/21, Fa0/22, Fa0/23, Fa0/24
1002  Fddi-default          active
1003  subinterface-default  active
1004  Fddi0101-default      active
1005  tunnel-default        active
S1-SITE1#
  
```


Points Possib le	Points Award ed
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6. You have been asked to use a subnet from the range 192.168.128.0/24 to support devices on VLAN 20. R1-SITE1 will perform InterVLAN routing between VLAN1 and VLAN20 at SITE1. Configure R1-SITE1 such that all devices on each VLAN can reach each other. Choose a subnet from the 192.168.128.0/24. Assume growth on the subnet up to no more than 100 hosts. Do not use the entire /24 for VLAN 20. Add a screenshot of the of the “show run | section interface” command from R1-SITE1.

The screenshot shows a network topology in Cisco Packet Tracer. R1-SITE1 is a central router connected to R2-SITE2 and S1-SITE2. R1-SITE1 has interfaces Gig0/0/0, Gig0/0/1, and Gig0/1. S1-SITE2 has interfaces Fa0/10 and Fa0/11. PC1, PC2, and PC3 are connected to the network. The configuration window for R1-SITE1 shows the following commands:

```

R1-SITE1#show run | section interface
interface GigabitEthernet0/0
ip address 192.168.128.1 255.255.255.252
duplex auto
speed auto
interface GigabitEthernet0/1
description SITE1_LAN
ip address 192.168.0.1 255.255.255.0
duplex auto
speed auto
interface GigabitEthernet0/1.20
 encapsulation dot1q 20
 ip address 192.168.128.1 255.255.255.120
 interface GigabitEthernet0/2
 ip 20 address
 duplex auto
 speed auto
 shutdown
 interface Vlan1
 no ip address
 shutdown
 R1-SITE1#

```

8

7. PCs in VLAN 1 are already receiving DHCP from R1. Provide DHCP from R1 for the new subnet created in question #6. Be sure to exclude any addresses that are statically configured on the subnet. Add a screenshot of the of the “show run | section ip dhcp” command from R1-SITE1.

Replace this shape w

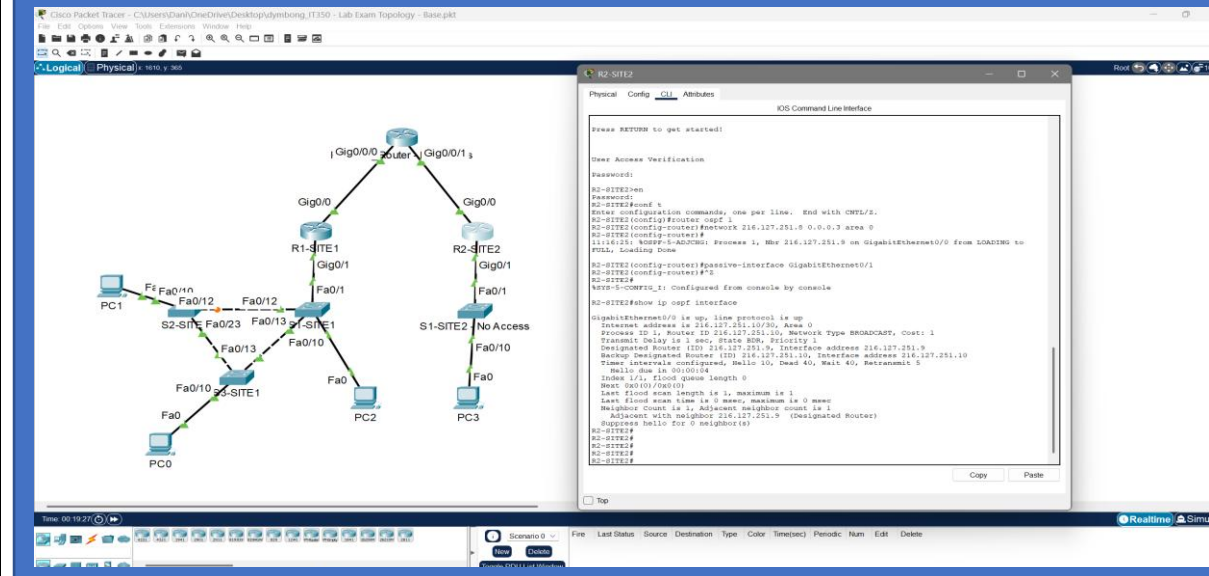
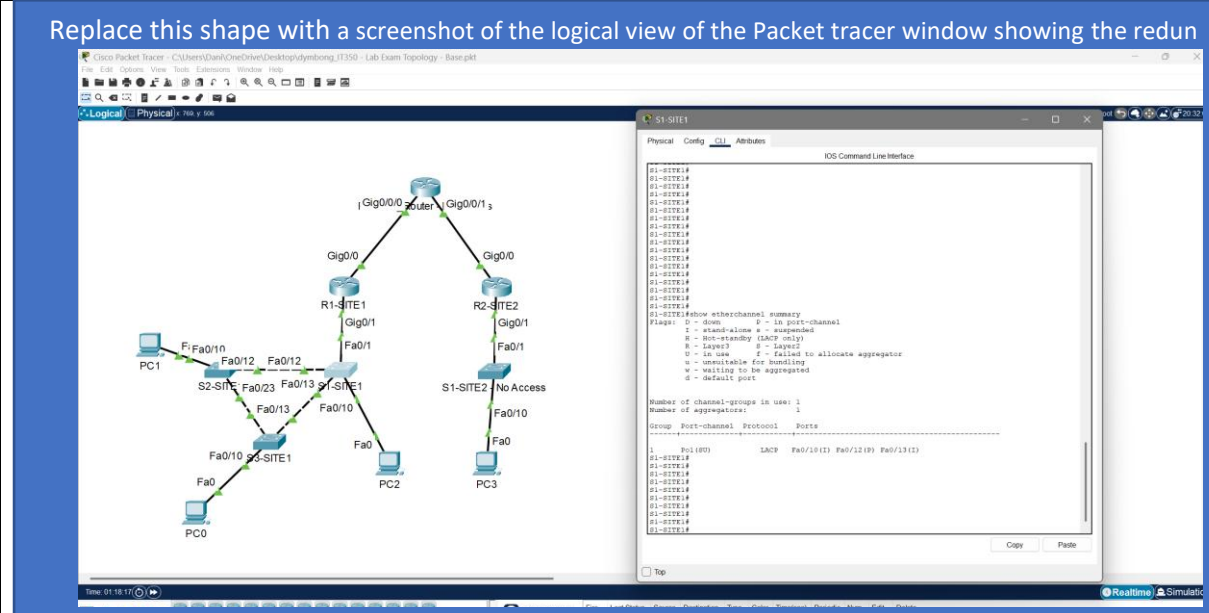
The screenshot shows a network topology in Cisco Packet Tracer. R1-SITE1 is a central router connected to R2-SITE2 and S1-SITE2. R1-SITE1 has interfaces Gig0/0/0, Gig0/0/1, and Gig0/1. S1-SITE2 has interfaces Fa0/10 and Fa0/11. PC1, PC2, and PC3 are connected to the network. The configuration window for R1-SITE1 shows the following commands:

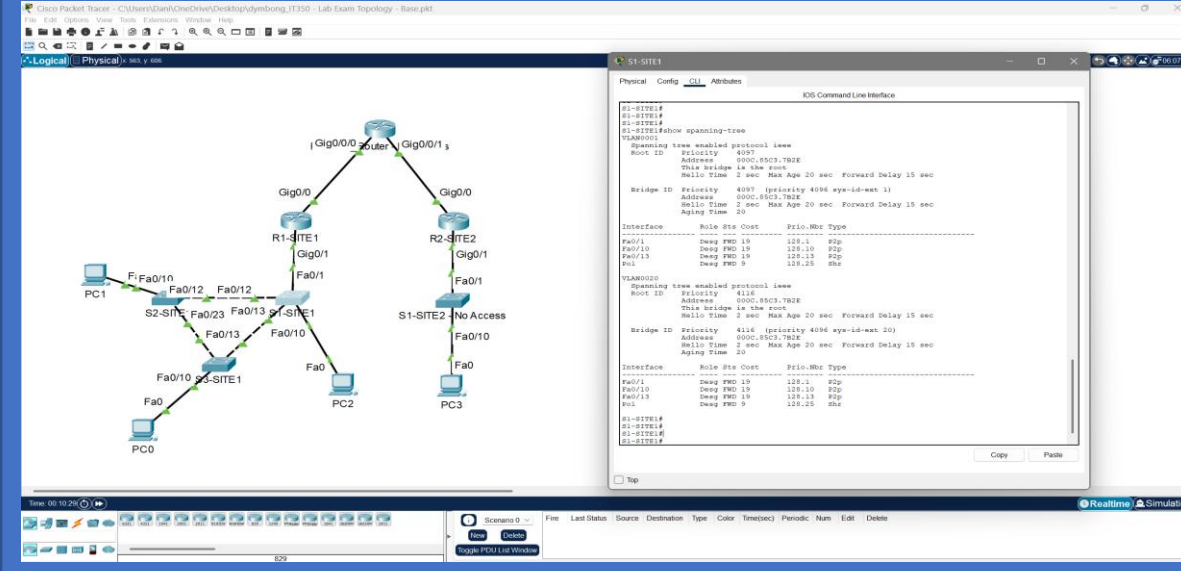
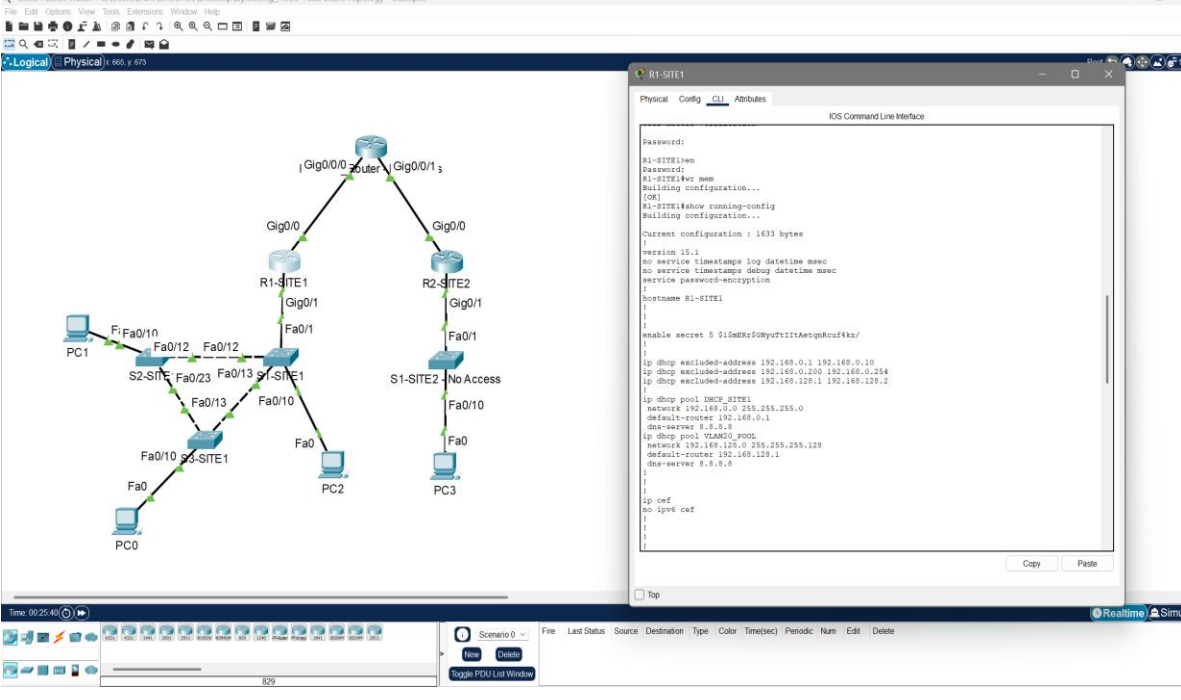
```

R1-SITE1#show run | section ip dhcp
ip dhcp excluded-address 192.168.0.1 192.168.0.10
ip dhcp excluded-address 192.168.0.200 192.168.0.254
ip dhcp excluded-address 192.168.128.1 192.168.128.2
ip dhcp pool DHCP_SITE1
 network 192.168.0.0 255.255.255.0
 default-router 192.168.0.1
 dns-server 8.8.8.8
 ip dhcp pool VLAN20_POOL
 network 192.168.128.0 255.255.255.120
 default-router 192.168.128.1
 dns-server 8.8.8.8
 R1-SITE1#

```

8

	Points Possib le	Points Award ed
10. R2-SITE2 is mostly configured but requires that OSPF be added so that it can form a neighbor relationship with Provider_Router. Provider_Router is already configured for OSPF in area 0. Adding the appropriate OSPF configuration to R2-SITE2 such that it advertises all connected subnets, forms a neighbor relationship with Provider_Router over GE0/0 and does not send any OSPF hello packets out the LAN interface (GE0/1). Add a screenshot of the output of the “show ip ospf interface” command from R2-SITE2.		
	9	
11. Add redundant links between S1-SITE1 and the other two switches at SITE1 and configure channel groups using the protocol of your choice. Add a screenshot of the logical view of the Packet tracer window showing the redundant links with a green status.		
Replace this shape with a screenshot of the logical view of the Packet tracer window showing the redun 	9	
12. Configure Spanning Tree Protocol such that all traffic flows towards S1-SITE1 as the root of the layer 2 network even if additional switches with a default configuration are added in the future. Add a screenshot of the output of the “show spanning-tree” command from S1-SITE1.		

	Points Possib le	Points Award ed
 The screenshot shows a Cisco Packet Tracer network diagram and the configuration for S1-SITE1. The network diagram on the left shows a central router connected to three sites: R1-SITE1, R2-SITE2, and S1-SITE1. R1-SITE1 is connected to S2-SITE1 and S3-SITE1. S2-SITE1 is connected to S3-SITE1. S3-SITE1 is connected to PC0. R2-SITE2 is connected to S1-SITE1. S1-SITE1 is connected to PC2 and PC3. The configuration window on the right shows the configuration for S1-SITE1, including the spanning tree enabled protocol, root ID, bridge ID, and interface configurations for Fa0/1, Fa0/10, and Fa0/13.	9	
13. Save your show run on your routers and switches with a “wr mem” command and copy them to a text file. Upload the ‘show run’ for all devices that you can access (R1-SITE1, R2-SITE2, S1-SITE1, S2-SITE1 and S3-SITE1) to Canvas. Also upload a copy of your saved PacketTracer file.		
 The screenshot shows a Cisco Packet Tracer network diagram and the configuration for R1-SITE1. The network diagram on the left is the same as the one in the previous screenshot. The configuration window on the right shows the configuration for R1-SITE1, including the password, hostname, version, service timestamps, service password-encryption, enable secret, and interface configurations for Fa0/1, Fa0/10, and Fa0/13.		

	Points Possible	Points Awarded
The screenshot displays a network topology in Cisco Packet Tracer. A central router connects two sites. On the left, Site 1 includes switches S1-SITE1 and S2-SITE1, connected to PCs PC0, PC1, and PC2. On the right, Site 2 includes switch S1-SITE2 connected to PC3. The configuration window for R2-SITE2 shows the following commands: <pre> R2-SITE2#conf t Enter configuration mode R2-SITE2(config)# R2-SITE2(config)#hostname R2-SITE2 R2-SITE2(config)#enable secret 5 \$1m\$Rr60yUttIItAetg8ouf4kz/ R2-SITE2(config)#ip cef R2-SITE2(config)#no ip vrf cef R2-SITE2(config)#license udi pid CISCO2911/K9 sn FTK1524MD7- R2-SITE2(config)#spanning-tree mode prst </pre>	10	10
This screenshot shows the same network topology as above. The configuration window for S1-SITE1 displays the following commands: <pre> S1-SITE1#show run Building configuration... Current configuration : 1531 bytes ! version 12.1 no service timestamps log datetime msec no service timestamps debug datetime msec service password-encryption ! hostname S1-SITE1 enable secret 5 \$1m\$Rr60yUttIItAetg8ouf4kz/ ! ! spanning-tree mode prst spanning-tree extend system-id spanning-tree vlan 1,20 priority 4096 interface Port-channel1 switchport mode trunk ! interface FastEthernet0/1 switchport trunk allowed vlan 1,20 switchport mode trunk ! interface FastEthernet0/2 ! interface FastEthernet0/3 ! interface FastEthernet0/4 ! interface FastEthernet0/5 ! interface FastEthernet0/6 ! interface FastEthernet0/7 ! interface FastEthernet0/8 ! interface FastEthernet0/9 --More-- </pre>	10	10

 Cisco Packet Tracer - C:\Users\Dan\OneDrive\Desktop\pymbong_IT350 - Lab Exam Topology - Base.pkt File Edit Options View Tools Extensions Window Help </
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